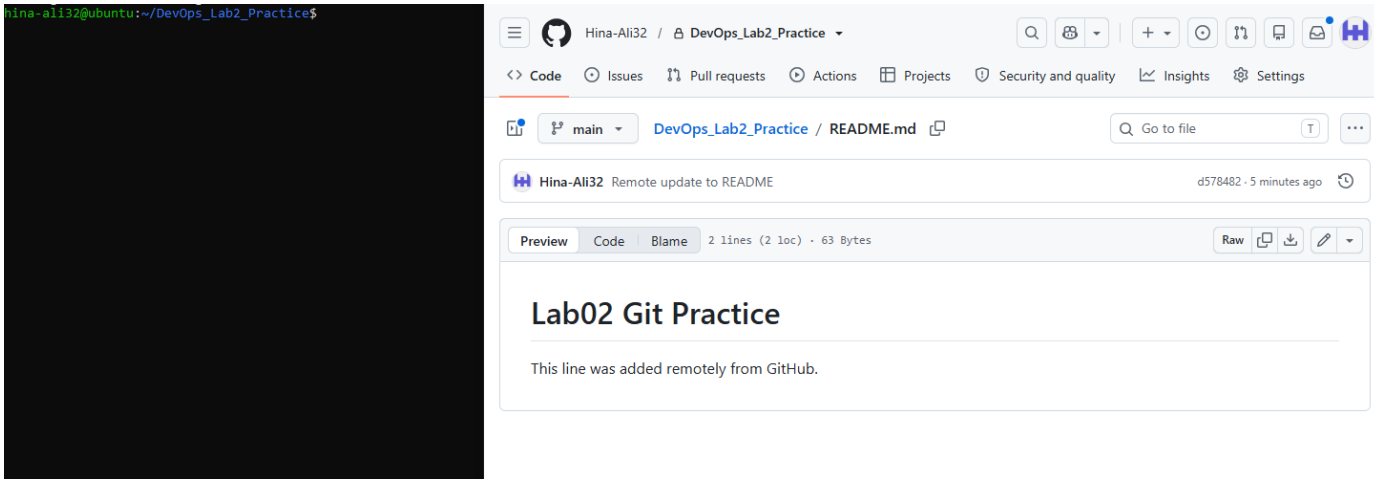
Fatima Jinnah Women University



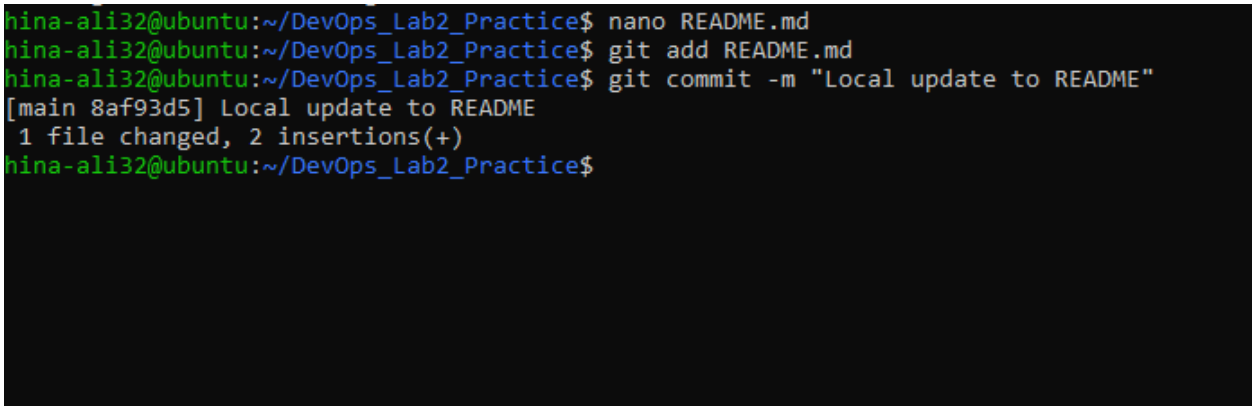
Department of Computer Science
DevOps Lab # 3

Name: Hina Ali
Registration no: 2023-BCS-040
Course code: CS-649
Section: BCS VII(B)
GitHub Username: Hina-Ali32

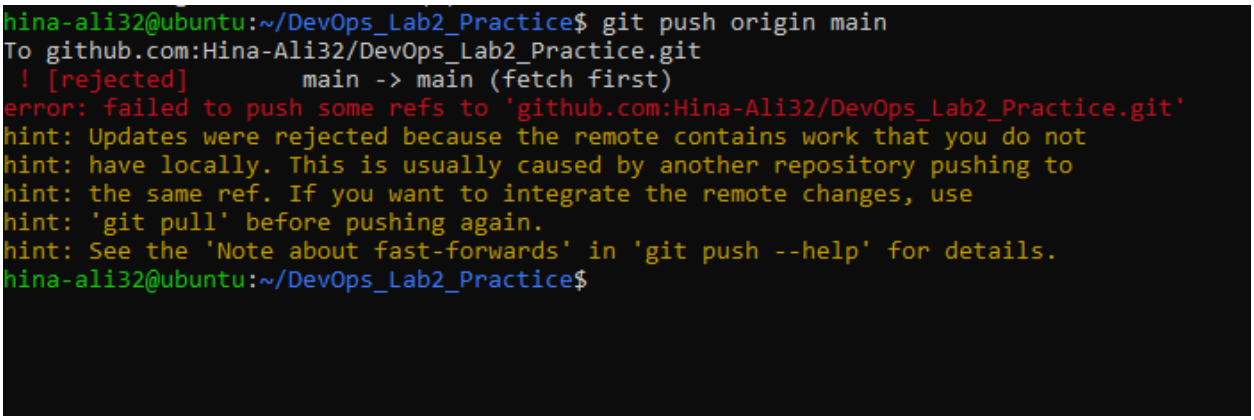
Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)



Shows README.md edited directly on GitHub, with the new line added remotely and committed as Remote update to README.



Shows the local README.md edit staged and committed as Local update to README, before any pull.



shows git push origin main rejected with fetch first, because the remote holds a commit that the local repository does not have.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Merge remote changes into local main"
[main 42afd09] Merge remote changes into local main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph -4
*   42afd09 (HEAD -> main) Merge remote changes into local main
|\
| * d578482 (origin/main, origin/HEAD) Remote update to README
* | 8af93d5 Local update to README
|/
* 91d4625 Add conflict_file.txt for Q3
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git pull --no-rebase origin main merging the remote commit into the local branch. The README conflict was resolved manually and the git log graph shows the merge commit joining both lines of history.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 691 bytes | 138.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   d578482..42afd09  main -> main
branch 'main' set up to track 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git push -u origin main succeeding after the merge, so local and remote are in sync.

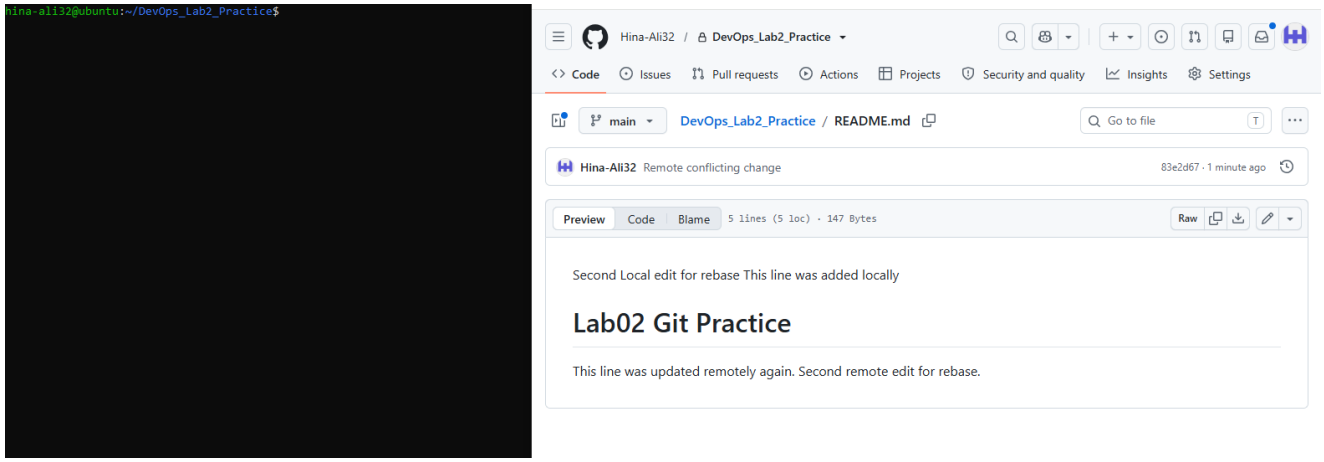
```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Local update 2 to README"
[main 0ed5e96] Local update 2 to README
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 976 bytes | 139.00 KiB/s, done.
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   42afd09..ad53072  main       -> origin/main
Successfully rebased and updated refs/heads/main.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph -4
* 02eb71e (HEAD -> main) Local update 2 to README
* ad53072 (origin/main, origin/HEAD) Remote update 2 to README
*   42afd09 Merge remote changes into local main
|\
| * d578482 Remote update to README
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git pull --rebase origin main finishing with Successfully rebased, and a straight-line log with the local commit replayed on top of the remote commit.

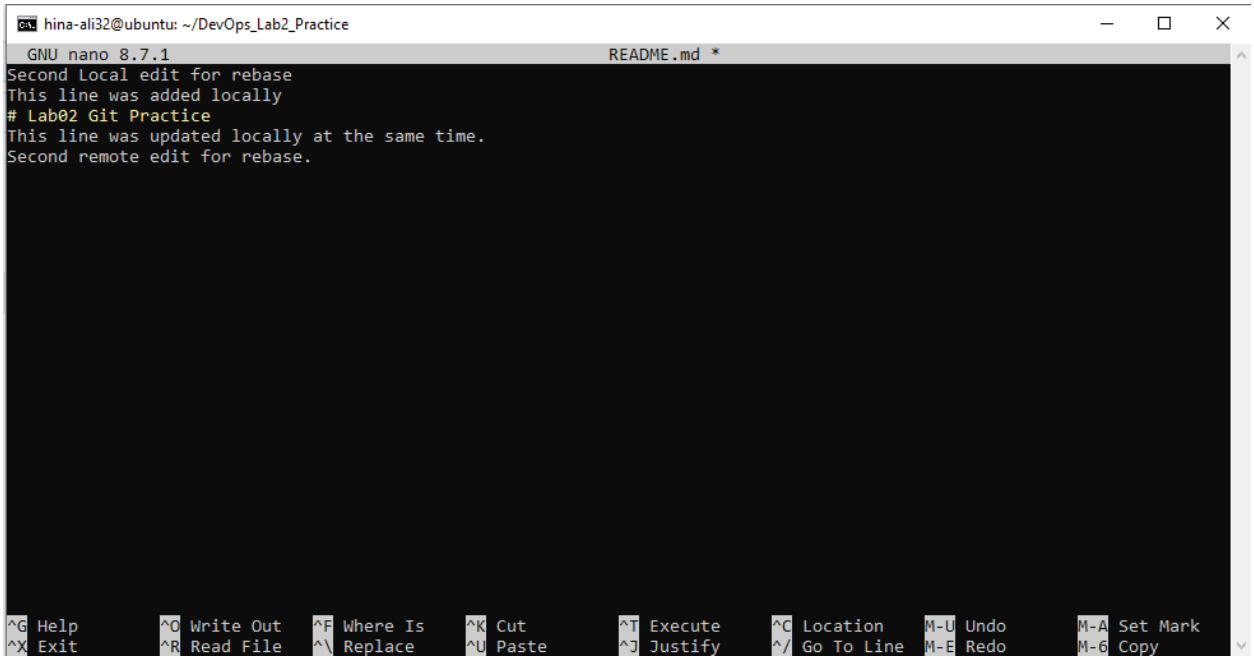
```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 348 bytes | 174.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   ad53072..02eb71e  main -> main
branch 'main' set up to track 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the rebased commit pushed successfully, with no merge commit created.

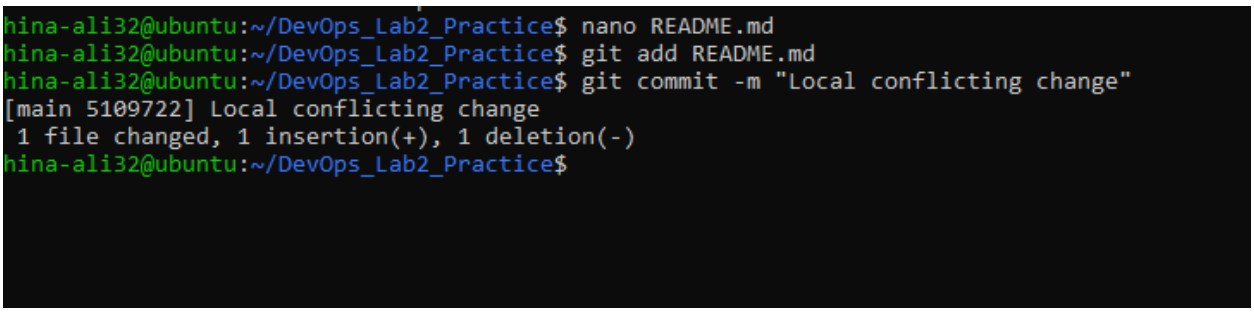
Task 2: Creating and Resolving Merge Conflicts Manually



Shows an existing README.md line changed on GitHub and committed as Remote conflicting change.



Shows the same line edited differently in the local editor, before saving.



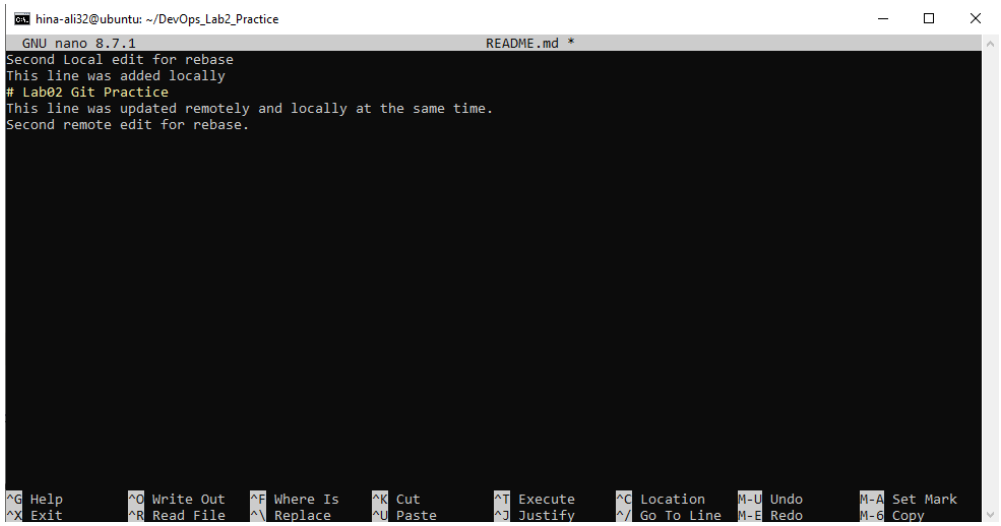
Shows the conflicting local edit staged and committed as Local conflicting change

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'github.com:Hina-Ali32/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

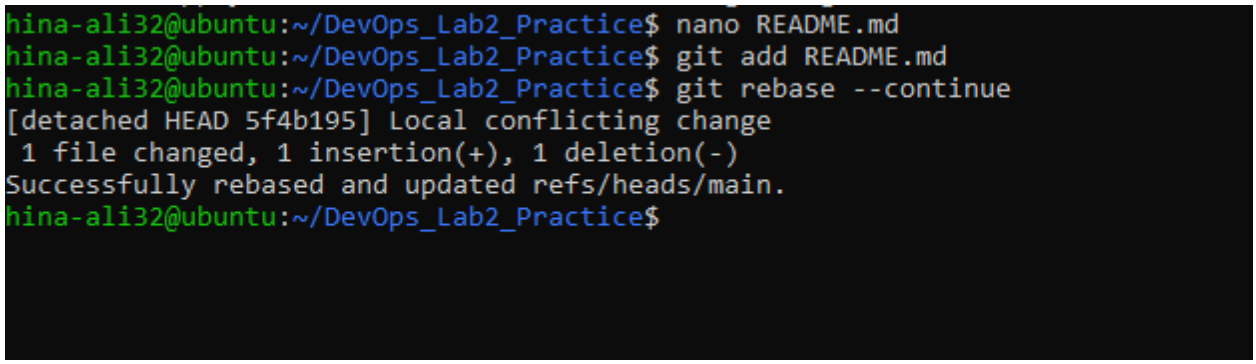
Shows the push rejected because the remote contains a commit the local branch lacks.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 5109722... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 5109722... # Local conflicting change
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

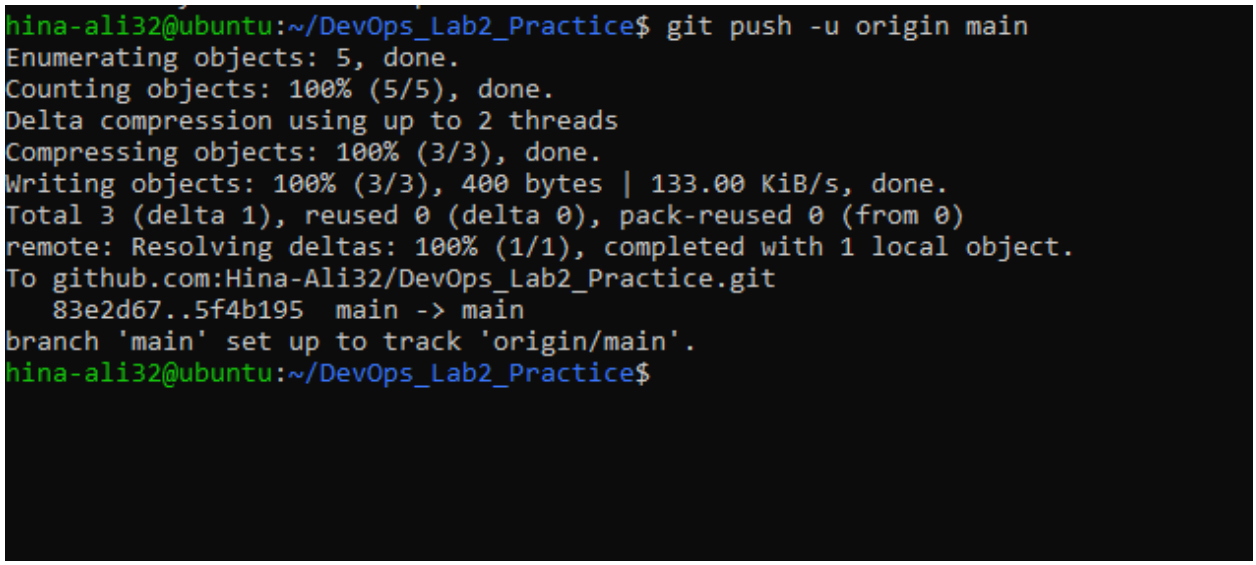
Shows git pull --rebase origin main stopping with CONFLICT (content) in README.md and Could not apply the local commit.



Shows README.md after the conflict markers were removed and both changes merged into a single line.



Shows git add README.md and git rebase --continue completing with Successfully rebased and updated refs/heads/main.



Shows the resolved commit pushed to origin main without errors.

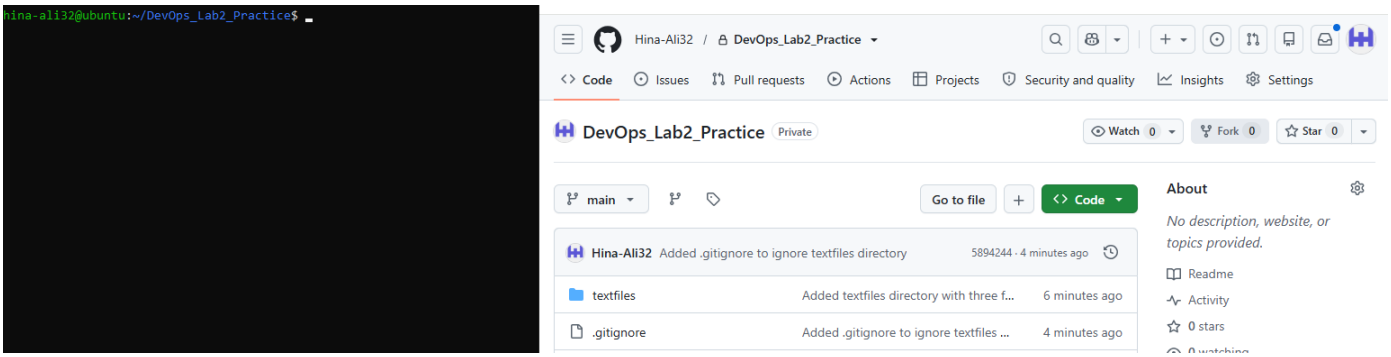
Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ mkdir textfiles
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "File A content" > textfiles/a.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "File B content" > textfiles/b.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "File C content" > textfiles/c.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added textfiles directory with three files"
[main 92d4182] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 449 bytes | 74.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   5f4b195..92d4182  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the textfiles folder with a.txt, b.txt and c.txt committed and pushed to main.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "textfiles/*" > .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added .gitignore to ignore textfiles directory"
[main 5894244] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 306 bytes | 43.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   92d4182..5894244  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

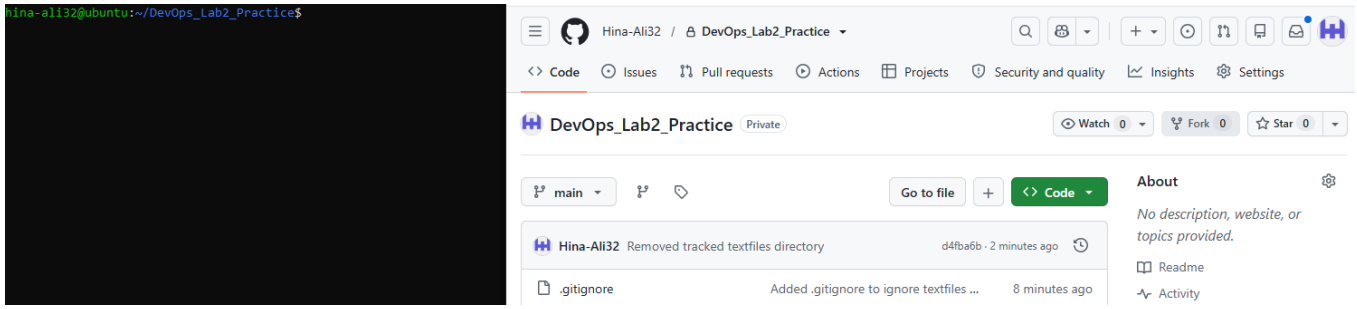
Shows .gitignore containing textfiles/* committed and pushed.



Shows the GitHub repository still listing the textfiles folder next to .gitignore, proving that .gitignore does not untrack files that are already committed.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Removed tracked textfiles directory"
[main d4fba6b] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 241 bytes | 34.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   5894244..d4fba6b  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git rm -r --cached textfiles removing the three files from tracking while keeping them on disk, followed by the commit and push.



Shows the GitHub repository after the push with the textfiles folder gone and .gitignore still present.

Task 4: Create Temporary Changes and Use git stash

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-branch
fatal: a branch named 'feature-branch' already exists
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Changes the README.md file"
[feature-branch 698fa4a] Changes the README.md file
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows feature-branch created and the README.md changes committed as Changes the README.md file.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git checkout main refused because local uncommitted changes to README.md would be overwritten.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git stash
Saved working directory and index state WIP on feature-branch: 698fa4a Changes the README.md file
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git stash saving the uncommitted changes as WIP on feature-branch.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git checkout main now succeeding after the stash.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows switching back to feature-branch.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git status reporting nothing to commit, working tree clean, because the edits are held in the stash.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (7b2fb459c9d4e1f556e55e7937caf56cb1924a91)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git stash pop restoring the stashed edits, with README.md listed as modified and unstaged on feature-branch.

Task 5: Checkout a Specific Commit Using git log

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
d4fba6b (HEAD -> main, origin/main, origin/HEAD) Removed tracked textfiles directory
5894244 Added .gitignore to ignore textfiles directory
92d4182 Added textfiles directory with three files
5f4b195 Local conflicting change
83e2d67 Remote conflicting change
02eb71e Local update 2 to README
ad53072 Remote update 2 to README
42afd09 Merge remote changes into local main
8af93d5 Local update to README
d578482 Remote update to README
91d4625 Add conflict_file.txt for Q3
d3a76bf Merge staging into main
dd265e2 (origin/staging, staging) Merge develop into staging
97352e4 (origin/develop, develop) Merge feature/q2-search into develop
d8dc8e3 (feature/q2-search) Add Q2 search feature
e6bf929 Merge q1-branch into main
ef74e6f (q1-branch) Q1: update notes.txt
26352fa (origin/collab) Update notes.txt
5fbd43d (collab) Add main account name and fun fact
122e4b1 Merge pull request #2 from Hina-Ali32/feature/code-review
1bf6423 (origin/feature/code-review, feature/code-review) Add description to review notes
4c7bae2 Add review notes
1862267 Merge staging into main
5bd3f37 Merge develop into staging
bac02ec Merge bugfix/login-error into develop
5111c76 Merge feature/user-profile into develop
0c0791d (bugfix/login-error) Fix login error
2c712ca (origin/feature/user-profile, feature/user-profile) Add user profile feature
5ffce89 Merge pull request #1 from Hina-Ali32/feature/db-connection
f0206f8 (origin/feature/db-connection, feature/db-connection) Add database connection notes
d5348b0 (origin/feature-1, feature-1) Add new function to main.py
3b56643 Update notes.txt
796354b Add notes.txt
8be1c58 (origin/bugfix/user-auth-error) Initial commit
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git log --oneline listing the commit history with short hashes.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout 02eb71e
Note: switching to '02eb71e'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 02eb71e Local update 2 to README
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git checkout with an older commit hash entering detached HEAD state at Local update 2 to README.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Previous HEAD position was 02eb71e Local update 2 to README
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git checkout main leaving the detached HEAD (Previous HEAD position was shown) and returning to the main branch.

Task 6: Resetting Commits (Soft vs Hard Reset)

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added test line"
[main efcc4dd] Added test line
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a new line added to README.md and committed as Added test line.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Second test commit"
[main 82382f4] Second test commit
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a second change committed as Second test commit.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
82382f4 (HEAD -> main) Second test commit
efcc4dd Added test line
d4fba6b (origin/main, origin/HEAD) Removed tracked textfiles directory
5894244 Added .gitignore to ignore textfiles directory
92d4182 Added textfiles directory with three files
5f4b195 Local conflicting change
83e2d67 Remote conflicting change
02eb71e Local update 2 to README
ad53072 Remote update 2 to README
42afd09 Merge remote changes into local main
8af93d5 Local update to README
d578482 Remote update to README
91d4625 Add conflict_file.txt for Q3
d3a76bf Merge staging into main
dd265e2 (origin/staging, staging) Merge develop into staging
97352e4 (origin/develop, develop) Merge feature/q2-search into develop
d8dc8e3 (feature/q2-search) Add Q2 search feature
e6bf929 Merge q1-branch into main
ef74e6f (q1-branch) Q1: update notes.txt
26352fa (origin/collab) Update notes.txt
5fbd43d (collab) Add main account name and fun fact
122e4b1 Merge pull request #2 from Hina-Ali32/feature/code-review
1bf6423 (origin/feature/code-review, feature/code-review) Add description to review notes
4c7bae2 Add review notes
1862267 Merge staging into main
5bd3f37 Merge develop into staging
bac02ec Merge bugfix/login-error into develop
5111c76 Merge feature/user-profile into develop
0c0791d (bugfix/login-error) Fix login error
2c712ca (origin/feature/user-profile, feature/user-profile) Add user profile feature
5ffce89 Merge pull request #1 from Hina-Ali32/feature/db-connection
f0206f8 (origin/feature/db-connection, feature/db-connection) Add database connection notes
d5348b0 (origin/feature-1, feature-1) Add new function to main.py
3b56643 Update notes.txt
796354b Add notes.txt
8be1c58 (origin/bugfix/user-auth-error) Initial commit
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git log --oneline with both new commits at the top of the history.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
Second Local edit for rebase
This line was added locally
# Lab02 Git Practice
This line was updated remotely and locally at the same time.
Second remote edit for rebase.
Reset test line 1
Reset test line 2.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows README.md containing both added lines before any reset.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git reset --soft HEAD~1
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git reset --soft HEAD~1 completing silently; the last commit is removed while its changes are kept.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
efcc4dd (HEAD -> main) Added test line
d4fba6b (origin/main, origin/HEAD) Removed tracked textfiles directory
5894244 Added .gitignore to ignore textfiles directory
92d4182 Added textfiles directory with three files
5f4b195 Local conflicting change
83e2d67 Remote conflicting change
02eb71e Local update 2 to README
ad53072 Remote update 2 to README
42afd09 Merge remote changes into local main
8af93d5 Local update to README
d578482 Remote update to README
91d4625 Add conflict_file.txt for Q3
d3a76bf Merge staging into main
dd265e2 (origin/staging, staging) Merge develop into staging
97352e4 (origin/develop, develop) Merge feature/q2-search into develop
d8dc8e3 (feature/q2-search) Add Q2 search feature
e6bf929 Merge q1-branch into main
ef74e6f (q1-branch) Q1: update notes.txt
26352fa (origin/collab) Update notes.txt
5fbd43d (collab) Add main account name and fun fact
122e4b1 Merge pull request #2 from Hina-Ali32/feature/code-review
1bf6423 (origin/feature/code-review, feature/code-review) Add description to review notes
4c7bae2 Add review notes
1862267 Merge staging into main
5bd3f37 Merge develop into staging
bac02ec Merge bugfix/login-error into develop
5111c76 Merge feature/user-profile into develop
0c0791d (bugfix/login-error) Fix login error
2c712ca (origin/feature/user-profile, feature/user-profile) Add user profile feature
5ffce89 Merge pull request #1 from Hina-Ali32/feature/db-connection
f0206f8 (origin/feature/db-connection, feature/db-connection) Add database connection notes
d5348b0 (origin/feature-1, feature-1) Add new function to main.py
3b56643 Update notes.txt
796354b Add notes.txt
8be1c58 (origin/bugfix/user-auth-error) Initial commit
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the log without Second test commit, with Added test line now at the top.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
Second Local edit for rebase
This line was added locally
# Lab02 Git Practice
This line was updated remotely and locally at the same time.
Second remote edit for rebase.
Reset test line 1
Reset test line 2.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows README.md still containing both lines, and git status listing the change as staged.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Second Test commit"
[main a117686] Second Test commit
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git commit recreating the second commit from the staged changes.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git reset --hard HEAD~1
HEAD is now at efcc4dd Added test line
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git reset --hard HEAD~1 reporting HEAD is now at Added test line.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
efcc4dd (HEAD -> main) Added test line
d4fba6b (origin/main, origin/HEAD) Removed tracked textfiles directory
5894244 Added .gitignore to ignore textfiles directory
92d4182 Added textfiles directory with three files
5f4b195 Local conflicting change
83e2d67 Remote conflicting change
02eb71e Local update 2 to README
ad53072 Remote update 2 to README
42afd09 Merge remote changes into local main
8af93d5 Local update to README
d578482 Remote update to README
91d4625 Add conflict_file.txt for Q3
d3a76bf Merge staging into main
dd265e2 (origin/staging, staging) Merge develop into staging
97352e4 (origin/develop, develop) Merge feature/q2-search into develop
d8dc8e3 (feature/q2-search) Add Q2 search feature
e6bf929 Merge q1-branch into main
ef74e6f (q1-branch) Q1: update notes.txt
26352fa (origin/collab) Update notes.txt
5fbd43d (collab) Add main account name and fun fact
122e4b1 Merge pull request #2 from Hina-Ali32/feature/code-review
1bf6423 (origin/feature/code-review, feature/code-review) Add description to review notes
4c7bae2 Add review notes
1862267 Merge staging into main
5bd3f37 Merge develop into staging
bac02ec Merge bugfix/login-error into develop
5111c76 Merge feature/user-profile into develop
0c0791d (bugfix/login-error) Fix login error
2c712ca (origin/feature/user-profile, feature/user-profile) Add user profile feature
5ffce89 Merge pull request #1 from Hina-Ali32/feature/db-connection
f0206f8 (origin/feature/db-connection, feature/db-connection) Add database connection notes
d5348b0 (origin/feature-1, feature-1) Add new function to main.py
3b56643 Update notes.txt
796354b Add notes.txt
8be1c58 (origin/bugfix/user-auth-error) Initial commit
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the log with the second commit permanently removed.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
Second Local edit for rebase
This line was added locally
# Lab02 Git Practice
This line was updated remotely and locally at the same time.
Second remote edit for rebase.
Reset test line 1
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows README.md without the second line, and git status reporting nothing to commit, working tree clean.

Task 7: Amending the Last Commit

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Amend test line 1." >> README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Fix log message"
[main 85fa0cb] Fix log message
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
85fa0cb (HEAD -> main) Fix log message
efcc4dd Added test line
d4fba6b (origin/main, origin/HEAD) Removed tracked textfiles directory
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a change committed as Fix log message, with the commit hash visible in the log.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ ls
README.md  conflict_file.txt  db_notes.md  login_fix.md  main.py  notes.txt  profile.md  q2_feature.md  review_notes.md  textfiles
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Amend test line 2." >> notes.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit --amend
[main 5af79dc] Fix log message and update notes
 Date: Sat Oct 17 16:41:19 2026 +0000
 2 files changed, 2 insertions(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
5af79dc (HEAD -> main) Fix log message and update notes
efcc4dd Added test line
d4fba6b (origin/main, origin/HEAD) Removed tracked textfiles directory
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git show --stat HEAD
commit 5af79dc07fb0cb1fb4af071df5ee9700a93cac99 (HEAD -> main)
Author: HinaAli <hina2015ali@gmail.com>
Date:   Sat Oct 17 16:41:19 2026 +0000

    Fix log message and update notes

 README.md | 1 +
 notes.txt | 1 +
 2 files changed, 2 insertions(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git commit --amend adding a second changed file to the same commit, with an updated message and a new hash. The log still has a single commit and git show --stat lists both files.

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "temporary text" >> temp.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added temporary text file"
[main 787918c] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 2 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (10/10), 885 bytes | 221.00 KiB/s, done.
Total 10 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 3 local objects.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   d4fba6b..787918c  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows temp.txt created, committed as Added temporary text file and pushed to origin main.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
787918c (HEAD -> main, origin/main, origin/HEAD) Added temporary text file
5af79dc Fix log message and update notes
efcc4dd Added test line
d4fba6b Removed tracked textfiles directory
5894244 Added .gitignore to ignore textfiles directory
92d4182 Added textfiles directory with three files
5f4b195 Local conflicting change
83e2d67 Remote conflicting change
02eb71e Local update 2 to README
ad53072 Remote update 2 to README
42afd09 Merge remote changes into local main
8af93d5 Local update to README
d578482 Remote update to README
91d4625 Add conflict_file.txt for Q3
d3a76bf Merge staging into main
dd265e2 (origin/staging, staging) Merge develop into staging
97352e4 (origin/develop, develop) Merge feature/q2-search into develop
d8dc8e3 (feature/q2-search) Add Q2 search feature
e6bf929 Merge q1-branch into main
ef74e6f (q1-branch) Q1: update notes.txt
26352fa (origin/collab) Update notes.txt
5fbd43d (collab) Add main account name and fun fact
122e4b1 Merge pull request #2 from Hina-Ali32/feature/code-review
1bf6423 (origin/feature/code-review, feature/code-review) Add description to review notes
4c7bae2 Add review notes
1862267 Merge staging into main
5bd3f37 Merge develop into staging
bac02ec Merge bugfix/login-error into develop
5111c76 Merge feature/user-profile into develop
0c0791d (bugfix/login-error) Fix login error
2c712ca (origin/feature/user-profile, feature/user-profile) Add user profile feature
5ffce89 Merge pull request #1 from Hina-Ali32/feature/db-connection
f0206f8 (origin/feature/db-connection, feature/db-connection) Add database connection notes
d5348b0 (origin/feature-1, feature-1) Add new function to main.py
3b56643 Update notes.txt
796354b Add notes.txt
8be1c58 (origin/bugfix/user-auth-error) Initial commit
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git revert 787918c
[main ef0013e] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
ef0013e (HEAD -> main) Revert "Added temporary text file"
787918c (origin/main, origin/HEAD) Added temporary text file
5af79dc Fix log message and update notes
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ ls
README.md  conflict_file.txt  db_notes.md  login_fix.md  main.py  notes.txt  profile.md  q2_feature.md  review_notes.md  textfiles
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git revert creating a new Revert commit that undoes temp.txt while the original commit stays in history.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 259 bytes | 129.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   787918c..ef0013e  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the revert commit pushed to origin main without rewriting history.

Task 9: Force Push (With Caution)

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b test-force
Switched to a new branch 'test-force'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/login-error
  collab
  develop
  feature-1
  feature-branch
  feature/code-review
  feature/db-connection
  feature/practice-reject
  feature/q2-search
  feature/user-profile
  main
  q1-branch
  q3-user1
  staging
* test-force
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the test-force branch created and checked out.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Force push test line." >> README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Force push test commit"
[test-force 4c711cb] Force push test commit
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a small change committed on test-force.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 312 bytes | 44.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/Hina-Ali32/DevOps_Lab2_Practice/pull/new/test-force
remote:
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 * [new branch]      test-force -> test-force
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows test-force pushed to GitHub as a new remote branch.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git reset --hard HEAD~1
HEAD is now at ef0013e Revert "Added temporary text file"
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git reset --hard HEAD~1 removing the last commit locally, so the local branch is now one commit behind the remote.

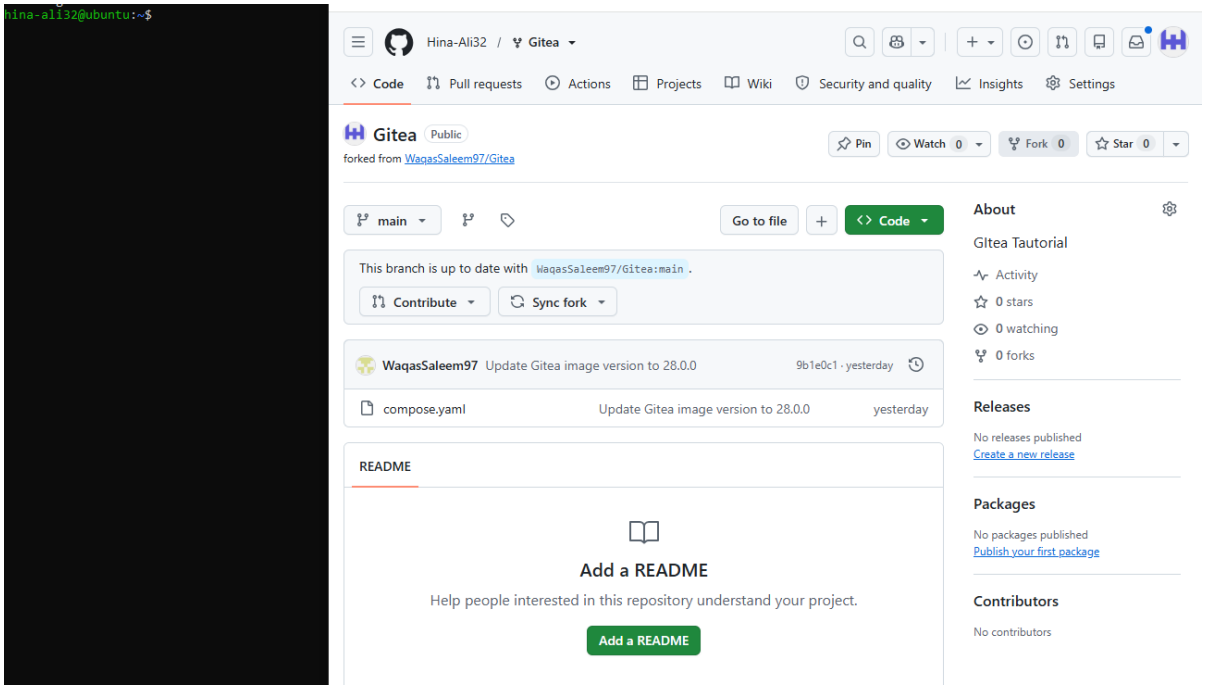
```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:Hina-Ali32/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a normal push rejected as non-fast-forward, because the local and remote histories have diverged.

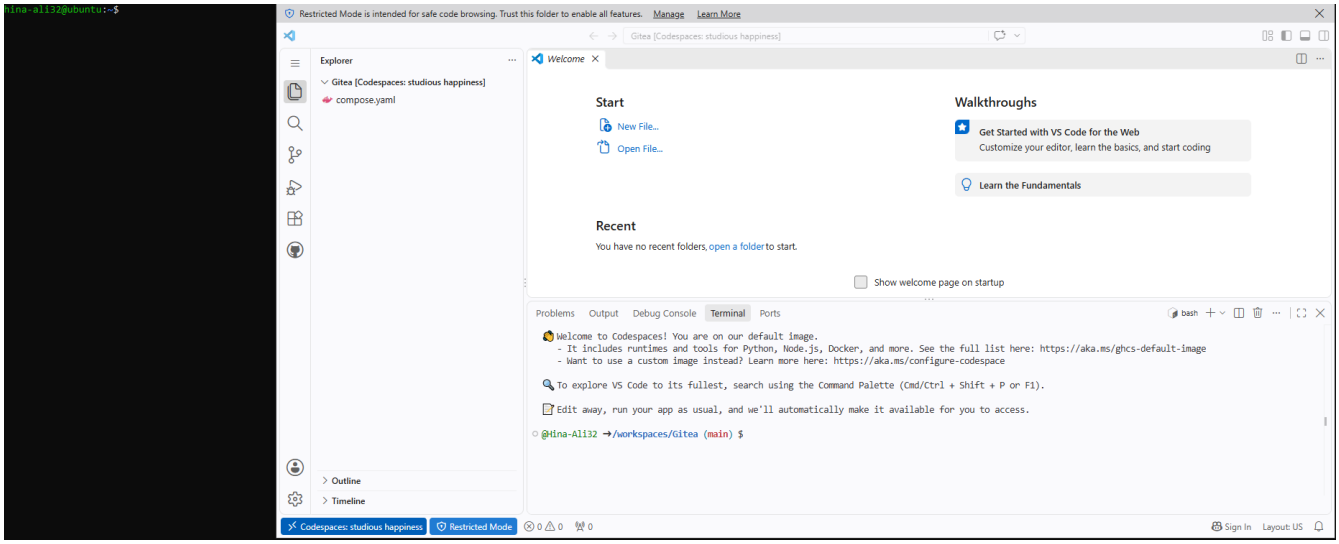
```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 + 4c711cb...ef0013e test-force -> test-force (forced update)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git push --force overwriting the remote test-force branch with the local history (forced update)

Task 10: Running Gitea in GitHub Codespaces via Docker Compose



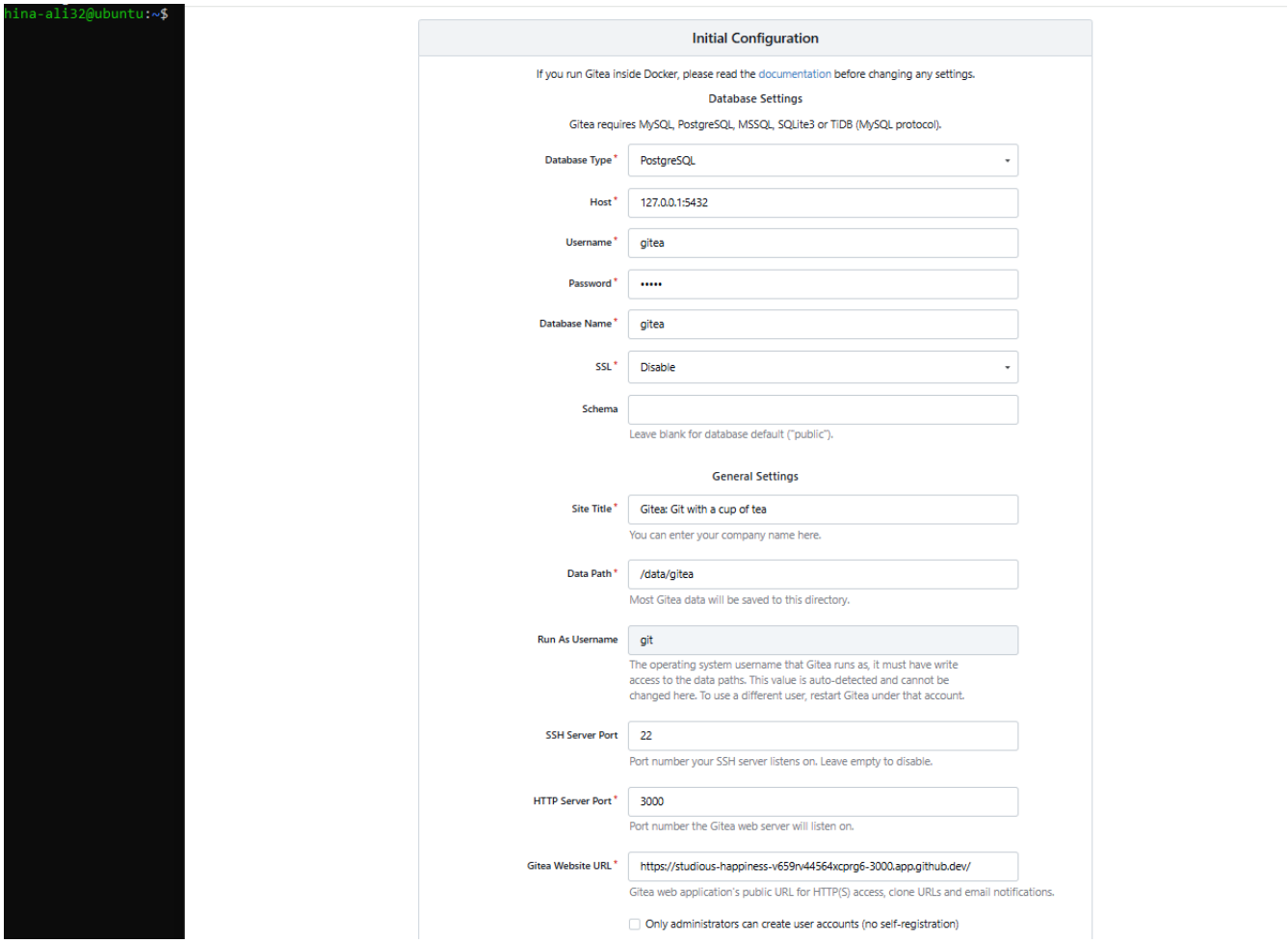
Shows the Gitea repository forked into the personal GitHub account, with the forked from line and 2 branches.

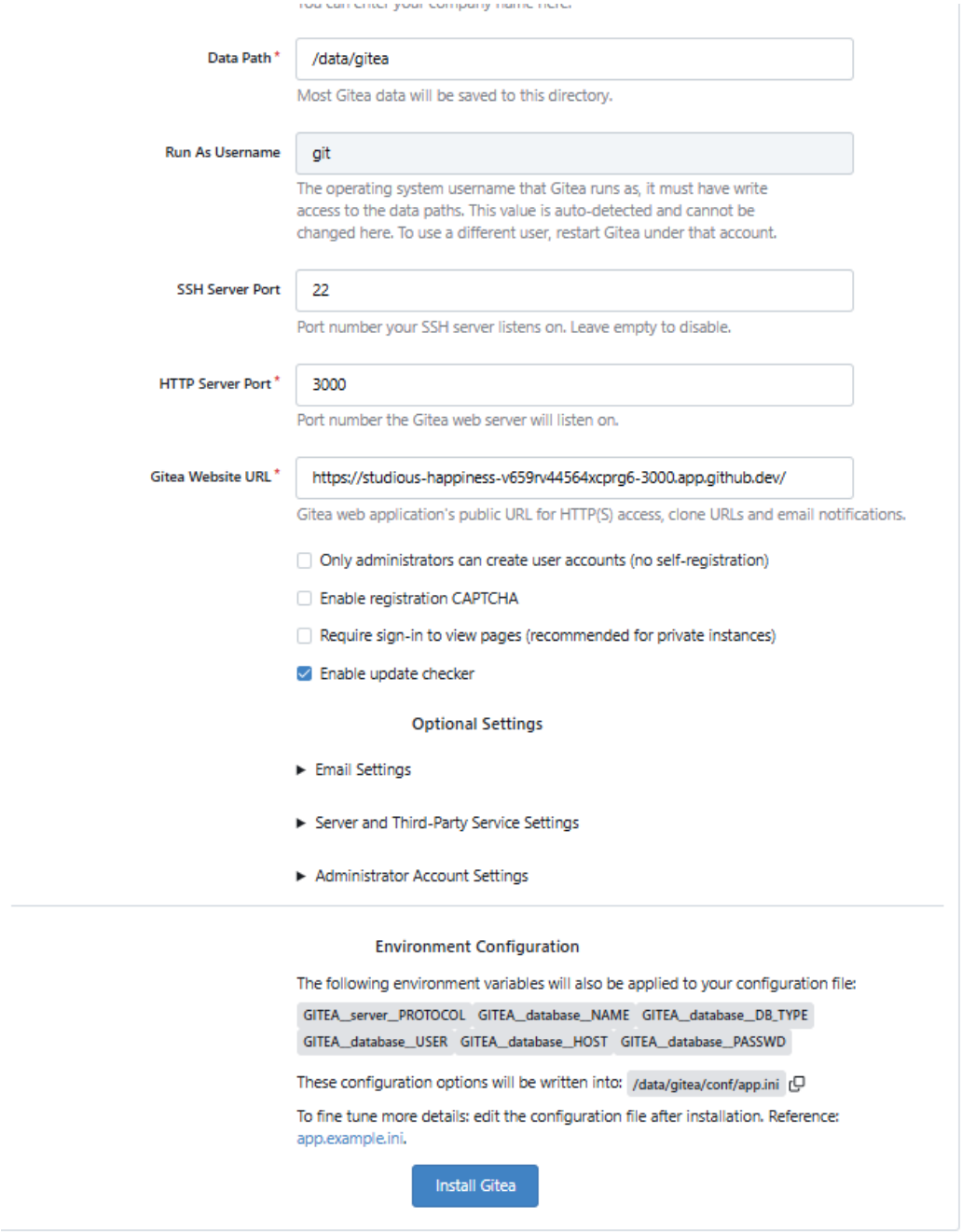


Shows the Codespace being created and started from the fork.

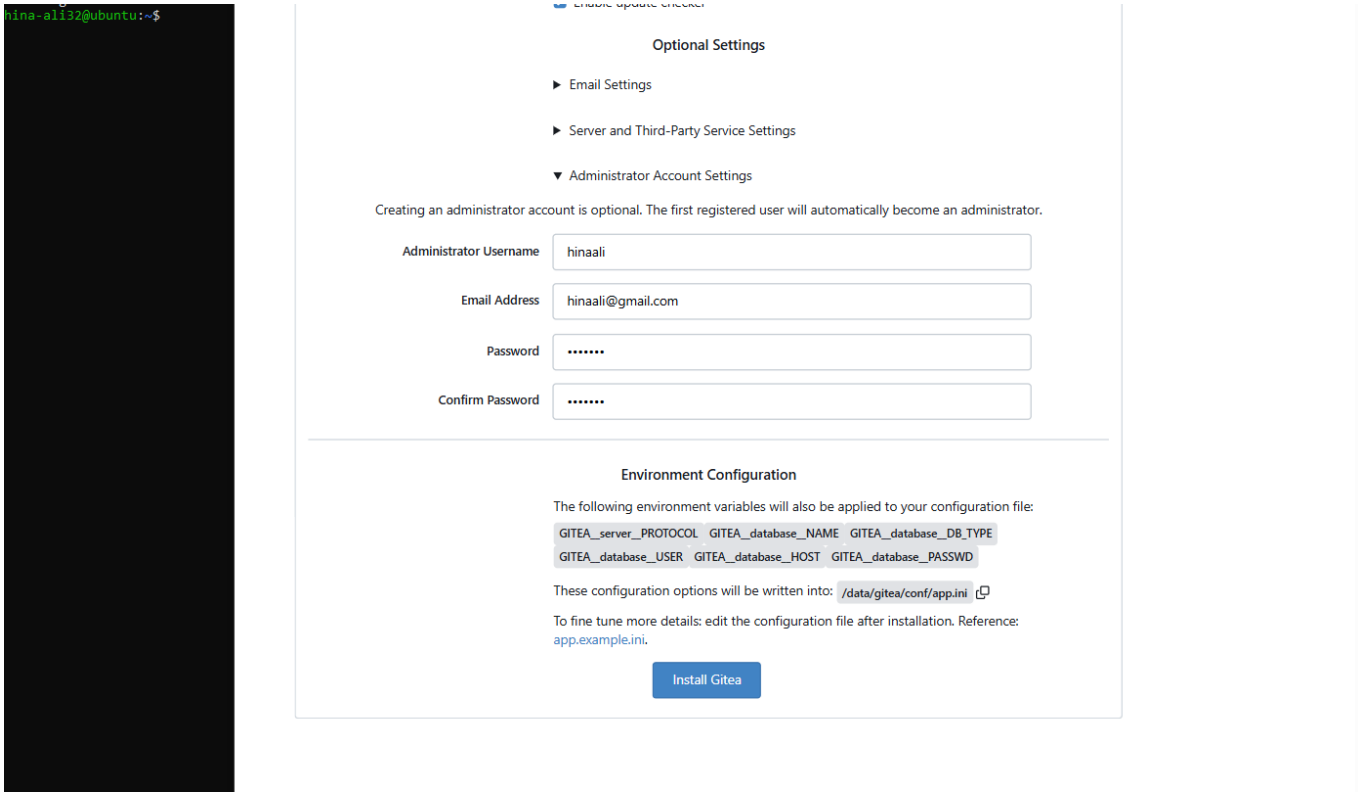
```
@Hina-Ali32 ➜/workspaces/Gitea (main) $ git checkout codespace
branch 'codespace' set up to track 'origin/codespace'.
Switched to a new branch 'codespace'
@Hina-Ali32 ➜/workspaces/Gitea (codespace) $ docker compose up -d
[+] up 24/25
 ✓ Image postgres:17-alpine Pulled
 ✓ Image gitea/gitea:28.0.0 Pulled
 ✓ Network webnet           Created
 ✓ Volume gitea_postgres    Created
 ✓ Volume gitea             Created
 ✓ Container gitea_db       Started
 ✓ Container gitea          Started
@Hina-Ali32 ➜/workspaces/Gitea (codespace) $ docker compose ps
NAME       IMAGE                COMMAND                  SERVICE   CREATED              STATUS              PORTS
gitea      gitea/gitea:28.0.0   "/usr/bin/entrypoint…"   gitea     About a minute ago   Up About a minute
gitea_db   postgres:17-alpine   "docker-entrypoint.s…"   db        About a minute ago   Up About a minute   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp, 5432/tcp
@Hina-Ali32 ➜/workspaces/Gitea (codespace) $
```

Shows docker compose up -d starting the Gitea containers, and docker compose ps listing them as running.

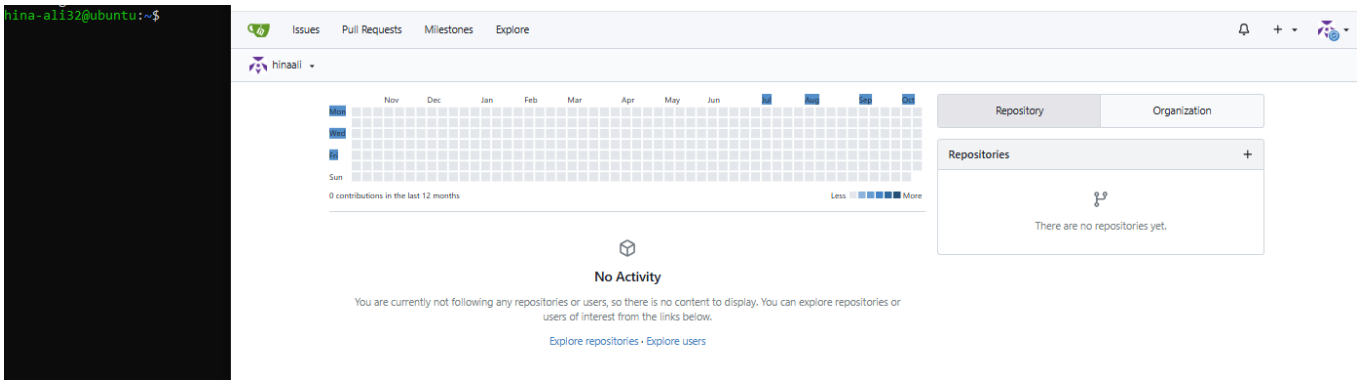




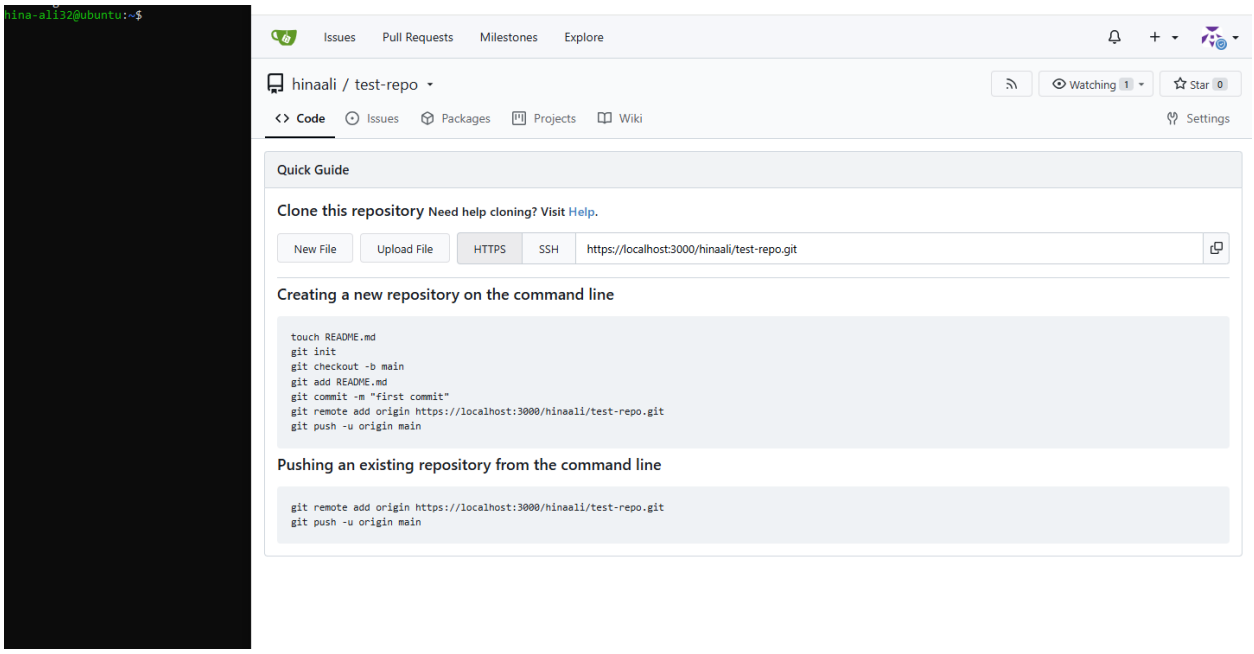
Shows the Gitea initial configuration page opened through the forwarded public port 3000.



Shows the Administrator Account Settings filled in before installation.

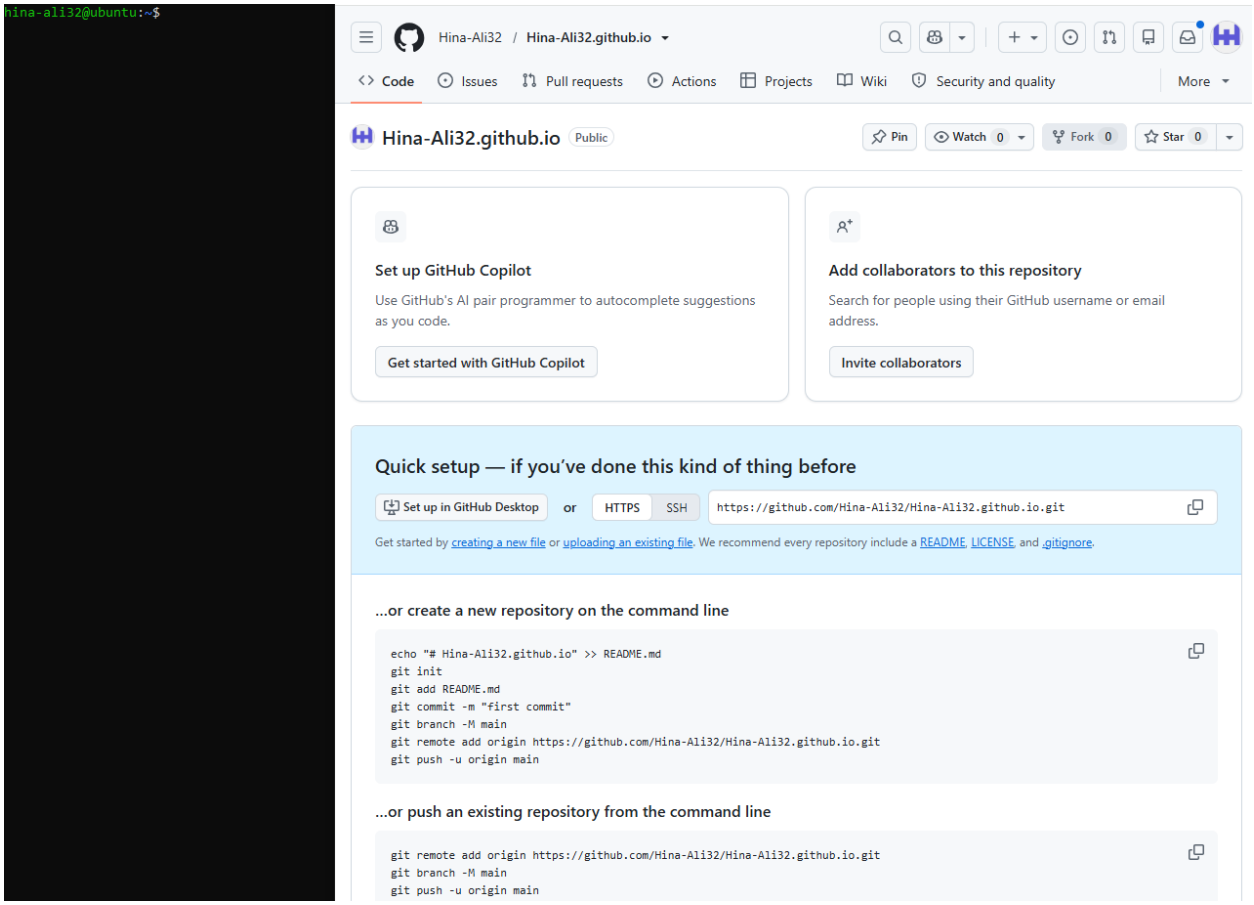


Shows the Gitea dashboard after installation and login with the admin account.



Shows the newly created repository inside Gitea.

Task 11: Creating a GitHub Pages Portfolio Site



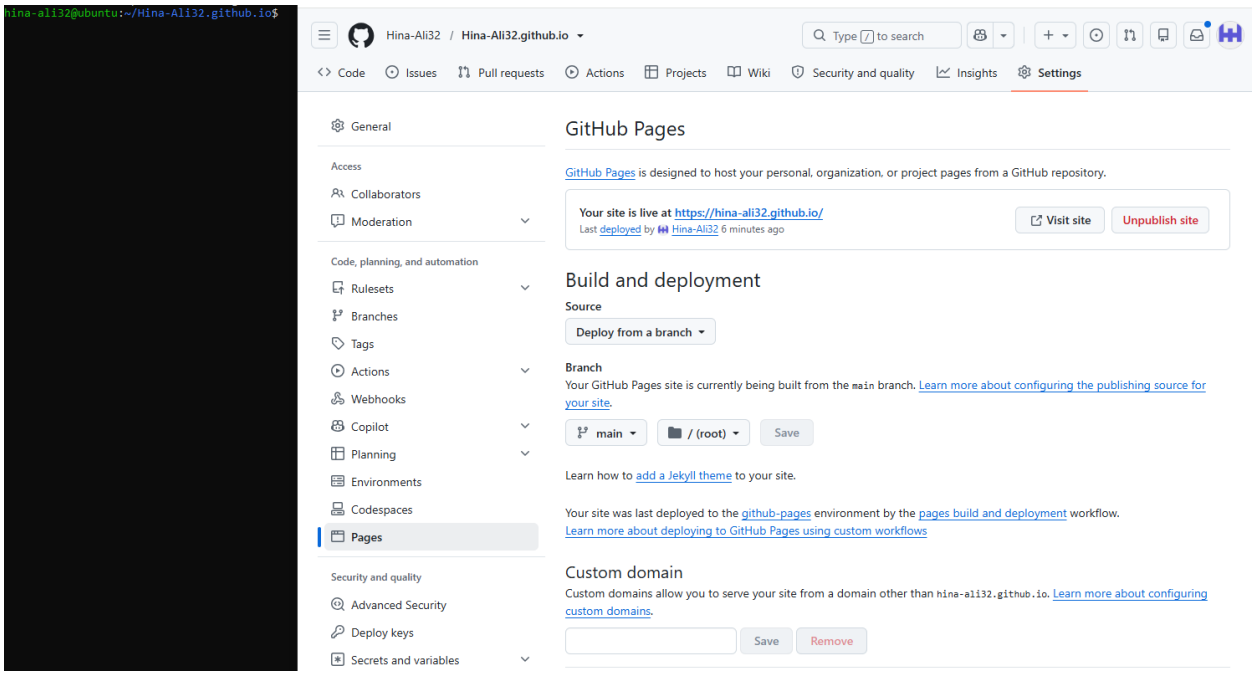
Shows the new empty public repository named after the username with .github.io.

```
hina-ali32@ubuntu:~/Hina-Ali32.github.io$ ls -la
total 16
drwxrwxr-x 2 hina-ali32 hina-ali32 4096 Oct 17 19:36 .
drwxr-x--- 8 hina-ali32 hina-ali32 4096 Oct 17 19:08 ..
-rw-rw-r-- 1 hina-ali32 hina-ali32 3855 Oct 17 19:27 index.html
-rw-rw-r-- 1 hina-ali32 hina-ali32 1779 Oct 17 19:08 style.css
hina-ali32@ubuntu:~/Hina-Ali32.github.io$
```

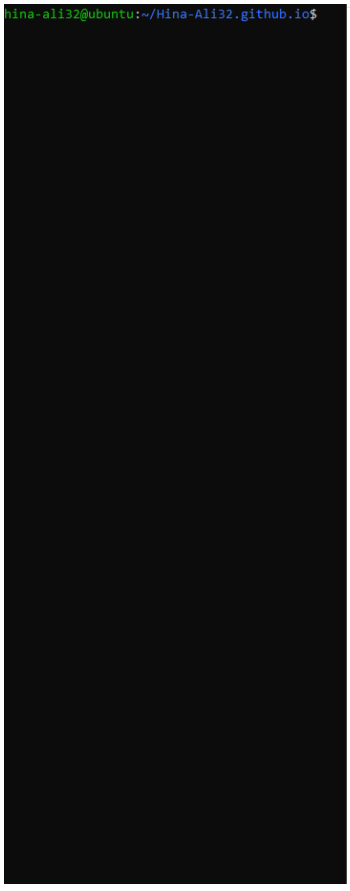
Shows the local site folder containing index.html and style.css.

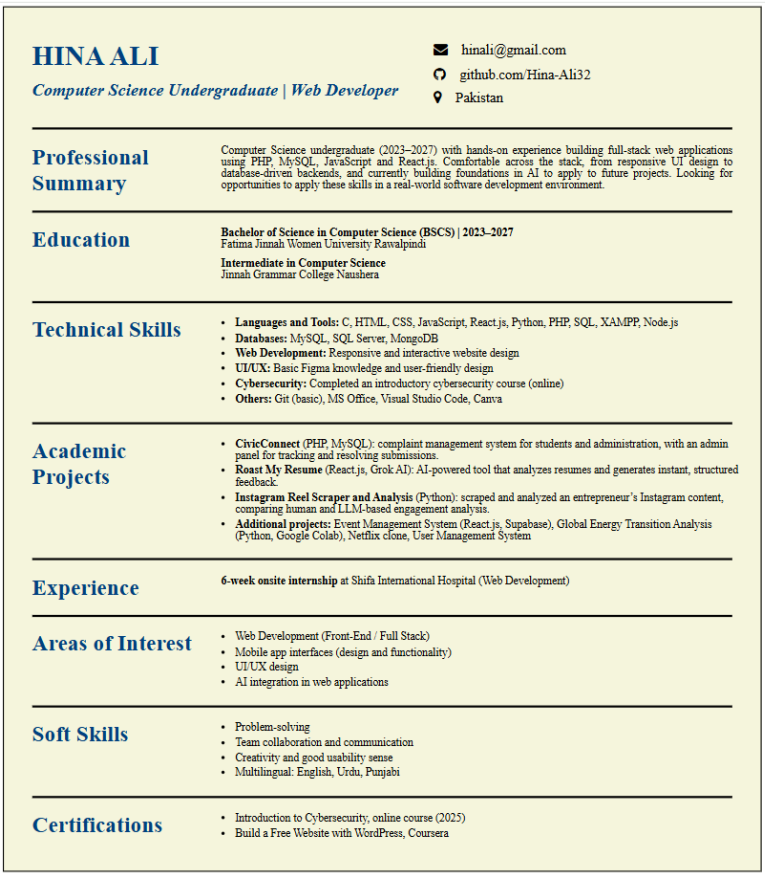
```
hina-ali32@ubuntu:~/Hina-Ali32.github.io$ git init -b main
Initialized empty Git repository in /home/hina-ali32/Hina-Ali32.github.io/.git/
hina-ali32@ubuntu:~/Hina-Ali32.github.io$ git remote add origin git@github.com:Hina-Ali32/Hina-Ali32.github.io.git
hina-ali32@ubuntu:~/Hina-Ali32.github.io$ git add .
hina-ali32@ubuntu:~/Hina-Ali32.github.io$ git commit -m "Add portfolio site for GitHub Pages"
[main (root-commit) 9adf730] Add portfolio site for GitHub Pages
 2 files changed, 243 insertions(+)
 create mode 100644 index.html
 create mode 100644 style.css
hina-ali32@ubuntu:~/Hina-Ali32.github.io$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 2.46 KiB | 419.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/Hina-Ali32.github.io.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
hina-ali32@ubuntu:~/Hina-Ali32.github.io$
```

Shows git init, git remote add, commit and push of the site files to main completing successfully.



Shows Settings, Pages confirming the site is published, with its live URL.





HINA ALI

Computer Science Undergraduate | Web Developer

hinali@gmail.com
github.com/Hina-Ali32
Pakistan

Professional Summary

Computer Science undergraduate (2023–2027) with hands-on experience building full-stack web applications using PHP, MySQL, JavaScript and React.js. Comfortable across the stack, from responsive UI design to database-driven backends, and currently building foundations in AI to apply to future projects. Looking for opportunities to apply these skills in a real-world software development environment.

Education

Bachelor of Science in Computer Science (BSCS) | 2023–2027
Fatima Jinnah Women University Rawalpindi

Intermediate in Computer Science
Jinnah Grammar College Naushera

Technical Skills

- **Languages and Tools:** C, HTML, CSS, JavaScript, React.js, Python, PHP, SQL, XAMPP, Node.js
- **Databases:** MySQL, SQL Server, MongoDB
- **Web Development:** Responsive and interactive website design
- **UI/UX:** Basic Figma knowledge and user-friendly design
- **Cybersecurity:** Completed an introductory cybersecurity course (online)
- **Others:** Git (basic), MS Office, Visual Studio Code, Canva

Academic Projects

- **CivicConnect** (PHP, MySQL): complaint management system for students and administration, with an admin panel for tracking and resolving submissions.
- **Roast My Resume** (React.js, Grok AI): AI-powered tool that analyzes resumes and generates instant, structured feedback.
- **Instagram Reel Scraper and Analysis** (Python): scraped and analyzed an entrepreneur's Instagram content, comparing human and LLM-based engagement analysis.
- **Additional projects:** Event Management System (React.js, Supabase), Global Energy Transition Analysis (Python, Google Colab), Netflix clone, User Management System

Experience

6-week onsite internship at Shifa International Hospital (Web Development)

Areas of Interest

- Web Development (Front-End / Full Stack)
- Mobile app interfaces (design and functionality)
- UI/UX design
- AI integration in web applications

Soft Skills

- Problem-solving
- Team collaboration and communication
- Creativity and good usability sense
- Multilingual: English, Urdu, Punjabi

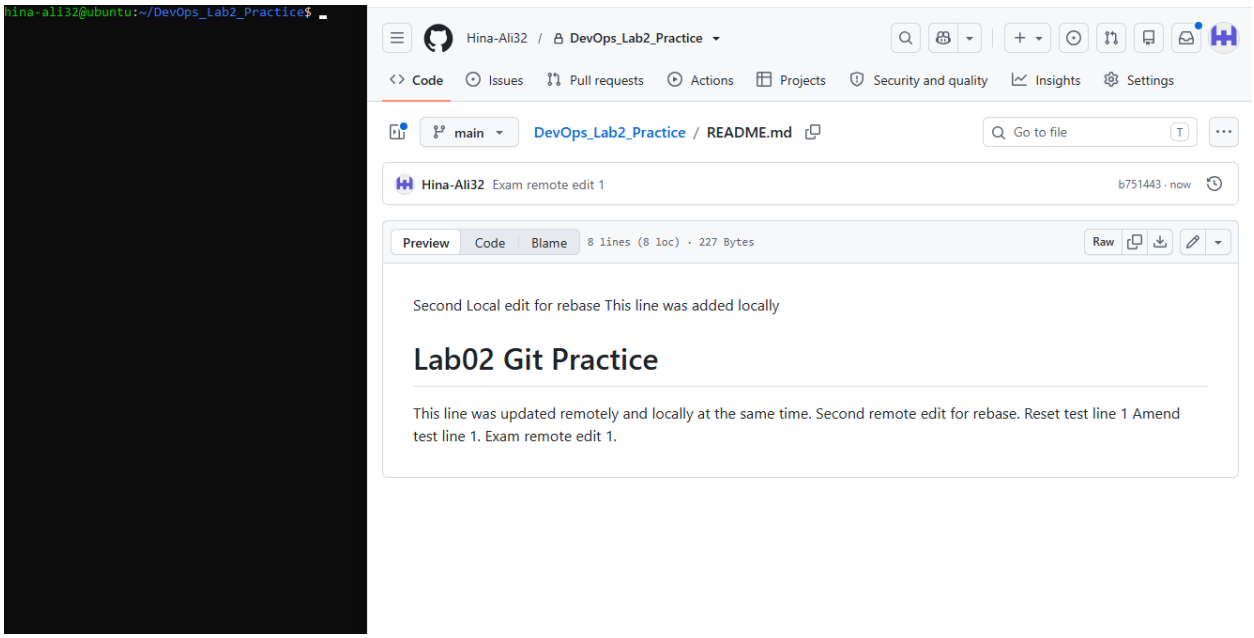
Certifications

- Introduction to Cybersecurity, online course (2025)
- Build a Free Website with WordPress, Coursera

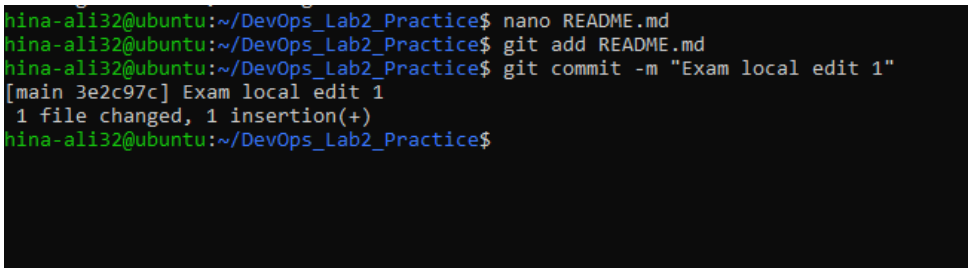
Shows the published CV page loading at its GitHub Pages address.

Exam Evaluation Questions

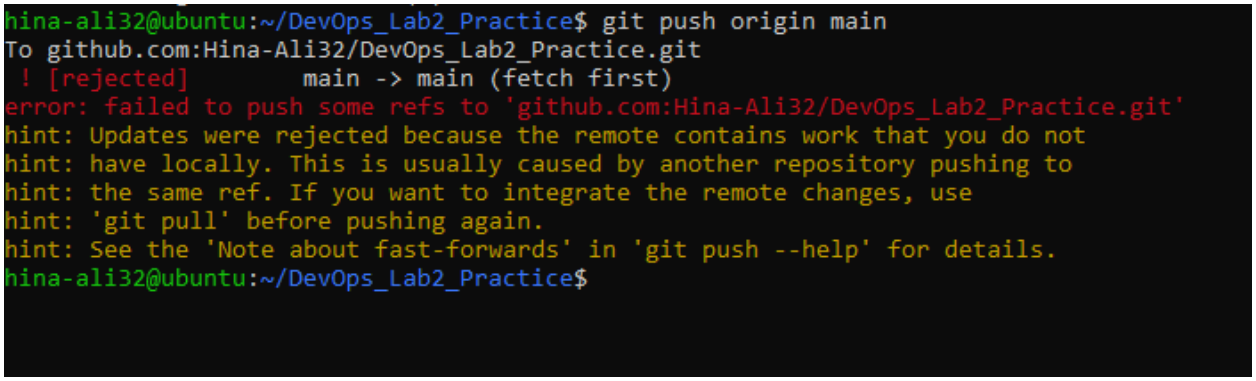
1. Local vs Remote Conflict Resolution



Shows README.md edited on GitHub and committed as Exam remote edit 1.



Shows a different local edit (at the top of the file) committed as Exam local edit 1, avoiding a conflict.



Shows the push rejected with fetch first, since the remote is ahead.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull --no-rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 947 bytes | 189.00 KiB/s, done.
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   ef0013e..b751443  main       -> origin/main
Auto-merging README.md
hint: Waiting for your editor to close the file...
```

```
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Merge remote changes"
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

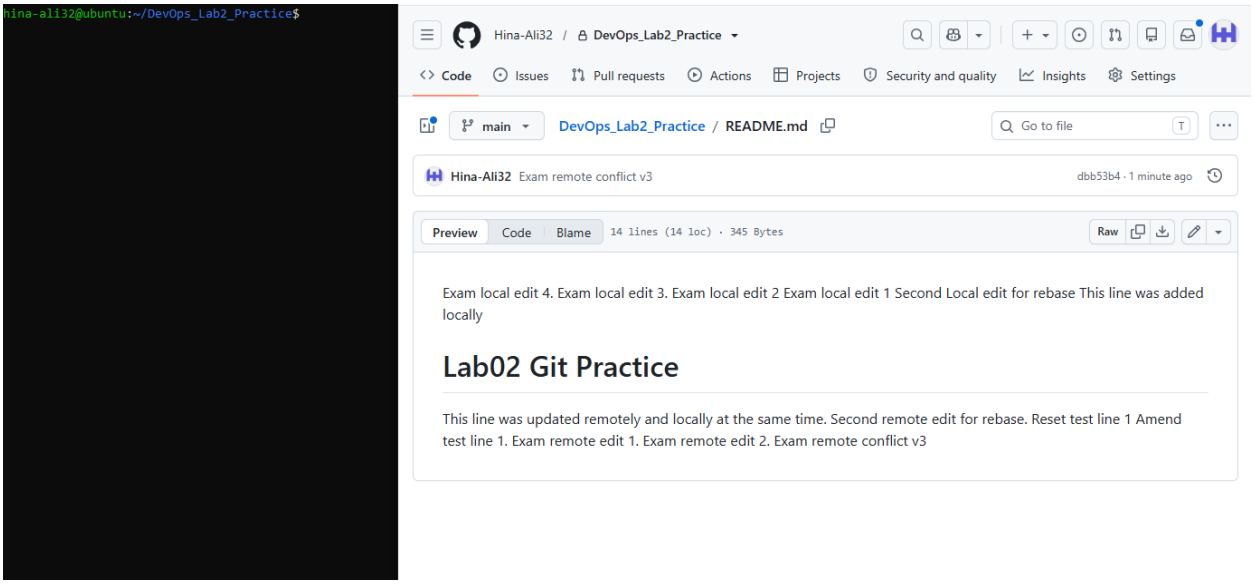
nothing to commit, working tree clean
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph -4
*   e4e2770 (HEAD -> main) Merge branch 'main' of github.com:Hina-Ali32/DevOps_Lab2_Practice
|\
| * b751443 (origin/main, origin/HEAD) Exam remote edit 1
* | 3e2c97c Exam local edit 1
|/
* ef0013e (origin/test-force, test-force) Revert "Added temporary text file"
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 609 bytes | 121.00 KiB/s, done.
Total 6 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 2 local objects.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   b751443..e4e2770  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git pull merging the remote commit, the log graph with the merge commit, and the successful push.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Exam local edit 4"
[main 26607dc] Exam local edit 4
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:Hina-Ali32/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.31 KiB | 267.00 KiB/s, done.
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   e5fdde9..9e3abec  main       -> origin/main
Successfully rebased and updated refs/heads/main.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph -4
* 2271e5f (HEAD -> main) Exam local edit 4
* 9e3abec (origin/main, origin/HEAD) Exam remote edit 4
* e5fdde9 Exam local edit 3
* 109f082 Exam local edit 2
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 310 bytes | 103.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   9e3abec..2271e5f  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the rejected push, git pull --rebase finishing with Successfully rebased, a straight-line log with the local commit on top, and the successful push.

2. Manual Merge Conflict Handling



Shows one specific README.md line changed on GitHub and committed.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Exam local conflict v3"
[main d122398] Exam local conflict v3
 1 file changed, 1 insertion(+), 1 deletion(-)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the same line changed differently locally and committed.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:Hina-Ali32/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the push rejected because the remote has commits that the local branch lacks.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 950 bytes | 950.00 KiB/s, done.
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   a1127f2..dbb53b4  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply d122398... Exam local conflict v3
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply d122398... # Exam local conflict v3
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git pull --rebase stopping with CONFLICT (content) in README.md.

```
hina-ali32@ubuntu: ~/DevOps_Lab2_Practice
  GNU nano 8.7.1                                README.md
Exam local edit 4.
Exam local edit 3.
Exam local edit 2
Exam local edit 1
Second Local edit for rebase
This line was added locally
# Lab02 Git Practice
This line was updated remotely and locally at the same time.
Second remote edit for rebase.
Reset test line 1
Amend test line 1.
Exam remote edit 1.
Exam remote edit 2.
Exam remote conflict v3
Exam local conflict v3

^G Help       ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

Shows README.md with the conflict markers removed and both edits merged into one line.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git rebase --continue
[detached HEAD 1f1c8fb] Exam local conflict v3
 1 file changed, 2 insertions(+)
Successfully rebased and updated refs/heads/main.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 322 bytes | 322.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   dbb53b4..1f1c8fb  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git add, git rebase --continue and git push completing, with the resolved commit pushed to main.

3. Managing Ignored and Tracked Files

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ mkdir DocFiles
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Doc A" > DocFiles/doc1.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Doc B" > DocFiles/doc2.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Doc C" > DocFiles/doc3.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ ls DocFiles
doc1.txt  doc2.txt  doc3.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the DocFiles folder created with several files inside.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added DocFiles directory"
[main ec23459] Added DocFiles directory
 3 files changed, 3 insertions(+)
 create mode 100644 DocFiles/doc1.txt
 create mode 100644 DocFiles/doc2.txt
 create mode 100644 DocFiles/doc3.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 414 bytes | 414.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   1f1c8fb..ec23459  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows DocFiles committed and pushed to GitHub.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "DocFiles/*" >> .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat .gitignore
textfiles/*
DocFiles/*
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

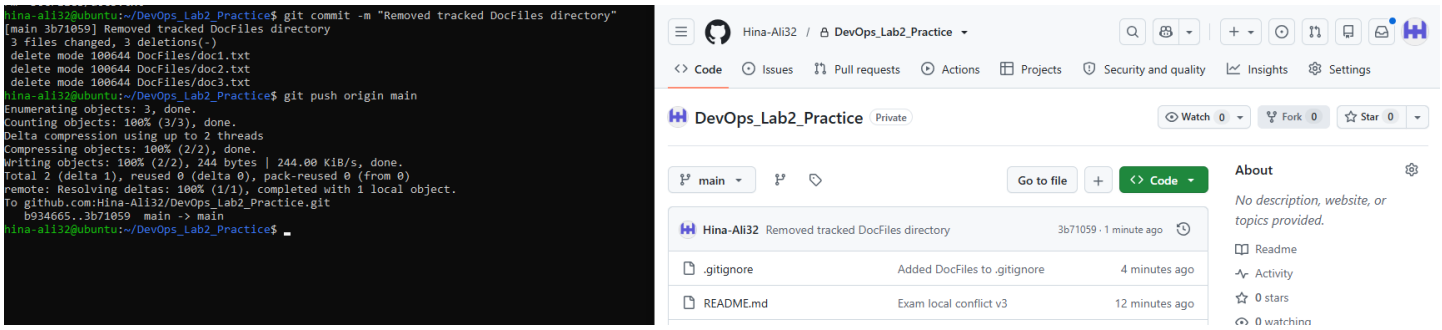
Shows DocFiles/* appended to .gitignore alongside the existing entry.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Added DocFiles to .gitignore"
[main b934665] Added DocFiles to .gitignore
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 303 bytes | 151.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   ec23459..b934665  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the .gitignore update committed and pushed.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git rm -r --cached DocFiles
rm 'DocFiles/doc1.txt'
rm 'DocFiles/doc2.txt'
rm 'DocFiles/doc3.txt'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git rm -r --cached DocFiles removing the files from tracking while keeping them locally.



Shows the GitHub repository after the push with the DocFiles folder no longer listed.

4. Commit History Manipulation and Recovery

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Q4 line 1." >> README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Q4 first commit"
[main 2894908] Q4 first commit
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a change committed as Q4 first commit.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Q4 line 2." >> README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add .
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Q4 second commit"
[main af06e29] Q4 second commit
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows another change committed as Q4 second commit.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -4
af06e29 (HEAD -> main) Q4 second commit
2894908 Q4 first commit
3b71059 (origin/main, origin/HEAD) Removed tracked DocFiles directory
b934665 Added DocFiles to .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git log --oneline with both Q4 commits at the top.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git reset --soft HEAD~1
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
2894908 (HEAD -> main) Q4 first commit
3b71059 (origin/main, origin/HEAD) Removed tracked DocFiles directory
b934665 Added DocFiles to .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ tail -3 README.md

Q4 line 1.
Q4 line 2.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git reset --soft HEAD~1 removing the second commit from history while both lines remain in README.md and are staged.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Q4 third commit"
[main 4e163e6] Q4 third commit
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
4e163e6 (HEAD -> main) Q4 third commit
2894908 Q4 first commit
3b71059 (origin/main, origin/HEAD) Removed tracked DocFiles directory
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a new commit created from the staged changes after the soft reset.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git reset --hard HEAD~1
HEAD is now at 2894908 Q4 first commit
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline -3
2894908 (HEAD -> main) Q4 first commit
3b71059 (origin/main, origin/HEAD) Removed tracked DocFiles directory
b934665 Added DocFiles to .gitignore
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ tail -3 README.md
Exam local conflict v3

Q4 line 1.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git reset --hard HEAD~1 removing the third commit and its changes, leaving a clean working tree.